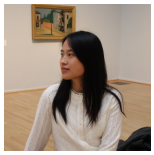
Elena Yan



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About

Elena Yan is a Ph.D. candidate at Mines Saint-Etienne, France and member of the Laboratory for Computer Science, Modeling and System Optimization, CNRS. She received her M.Sc. and B.Sc. degrees in Computer Science and Engineering at the University of Bologna, Italy. Her main research interests lie in multi-agent systems and engineering methodologies to design regulated, adaptive, and trustworthy systems. Particularly, her current research centers on designing an adaptive regulation system for multi-agent systems to cope with changing conditions, functional, and non-functional requirements in a cyber-physical social system.

Education

MINES Saint-Étienne, Saint-Étienne, France *Nov 2023 – Oct 2026*
Ph.D. in Computer Science

- *Thesis:* Self-Adaptive Regulation Mechanisms for a Trustworthy and Sustainable Industry of the Future
- *Supervisors:* Olivier Boissier, Jaime S. Sichman, and Luis G. Nardin
- *Keywords:* Multiagent Systems, Normative Systems, Responsible AI, Industry of the Future
- *Funding:* ANR/FAPESP Normative Artificial Intelligence for regulating MANufacturing - NAIMAN Project (ANR-22-CE23-0018-01, FAPESP 2022/03454-1)

University of Bologna, Campus of Cesena, Italy *Sep 2021 – Oct 2023*
Master's Degree in Computer Science and Engineering

- *Graduation grade:* 110/110 with Honors
- *Average grade:* 29,60/30
- *Thesis:* A Multi-Level Explainability Framework for BDI Multi-Agent Systems amslaurea.unibo.it/29644
- *Supervisors:* Alessandro Ricci, Jomi F. Hübner, and Samuele Burattini
- *Keywords:* Explainable Agency, Explainability, Belief-Desire-Intention Agent

University of Bologna, Campus of Cesena, Italy *Sep 2018 – Jul 2021*
Bachelor's Degree in Computer Science and Engineering

- *Thesis:* Telemedicine and Wearable Computing to Support Healthcare Professionals in Stroke Diagnosis: the TeleStroke Project as a Case Study amslaurea.unibo.it/23876 (Italian)
- *Supervisors:* Alessandro Ricci and Angelo Croatti
- *Keywords:* Wearable Computing, Smartglasses, Telemedicine

Experience

Visiting Ph.D. Student *Florianópolis, Brazil*
Federal University of Santa Catarina - Department of Automation and Systems Engineering *Aug 2025*

- *Supervisor:* Jomi F. Hübner

Visiting Ph.D. Student *São Paulo, Brazil*
University of São Paulo - Escola Politécnica, Laboratório de Técnicas Inteligentes *Jul 2025 - Oct 2025*

- *Supervisor:* Jaime S. Sichman

Ph.D. Student *Saint Etienne, France*
MINES Saint-Étienne, Saint-Étienne, France *Nov 2023 - Oct 2026*

- *Supervisors:* Olivier Boissier, Jaime S. Sichman, and Luis G. Nardin

Research Intern *Cesena, Italy*
Pervasive Software Lab - PSLAB, University of Bologna *Mar 2021 – May 2021*

- Design and development of software components for wearable technology systems applied in healthcare.
- *Supervisors:* Alessandro Ricci and Angelo Croatti

Intern *Forlì, Italy*
Municipality of Forlì - Traffic Office *Jan 2017 – Feb 2017*

Intern *Forlì, Italy*
Arte e Ricamo S.r.l. *May 2016 – Jun 2016*

Activities

Teaching

- 2025, 2026. Integrating and Engineering Intelligent Systems, 3rd year graduate course in Science and Engineering (Ingénieur Civil des Mines), MINES Saint-Étienne, Saint-Étienne.
- 2024, 2025. Multi-Agent Coordination, 2nd year Master Cyber-Physical Social Systems, MINES Saint-Étienne, Saint-Étienne.
- 2024, 2025. Object-Oriented Programming, 1st year, graduate course in Science and Engineering (Ingénieur Civil des Mines), MINES Saint-Étienne, Saint-Étienne.
- 2024, 2025. Industrial Risk and Accident Project - Modern Languages, 1st year graduate course in Science and Engineering (Ingénieur Civil des Mines), MINES Saint-Étienne, Saint-Étienne.

Supervision

- Amna (2026). An explainability framework for BDI normative agents. Co-supervision with Luis G. Nardin. Master Research Internship.

Scientific Events

- Co-Organizer. Summer School on Technologies for Trust, Interoperability, Autonomy, and Resilience in Industry 4.0 (AI4Industry 2026)
- Co-Chair. International Workshop on Coordination, Organizations, Institutions, Norms and Ethics for Governance of Multi-Agent Systems (COINE@AAMAS 2026)
- Co-Organizer. Summer School on Technologies for Trust, Interoperability, Autonomy, and Resilience in Industry 4.0 (AI4Industry 2025)

Program Committee

- PC Member of Workshop-School on Agents, Environments, and Applications WESAAC@BRACIS 2026
- PC Member of Italian National Workshop "Workshop From Objects to Agents" WOA 2026
- PC Member of Coordination, Organizations, Institutions, Norms and Ethics for Governance of Multi-Agent Systems COINE@AAMAS 2026
- PC Member of Autonomous Agents and Multi-Agent Systems AAMAS 2026
- PC Member of Italian National Workshop "Workshop From Objects to Agents" WOA 2025
- Sub-reviewer in 27th European Conference on Artificial Intelligence ECAI 2024

Presentations

- A Normative Model to Adapt the Regulation Management of Multi-Agent Systems. *SeReCo Spring Workshop*, Nuremberg, Germany
- An Adaptive Regulation Management System for Multi-Agent Systems: A Normative Approach. *MAFTEC*, Lyon France.
- A Regulation Adaptation Model for Multi-Agent Systems. *ECAI 2025*, Bologna, Italy.
- Towards an Ontology for Uniform Representations of Agent State for Heterogeneous Inter-Agent Explanations *HyperAgents @ ECAI 2025*, Bologna, Italy.
- Self-Adaptive Regulation Mechanisms for a Trustworthy and Sustainable Industry of the Future. *ECAI 2025 - Doctoral Consortium*, Bologna, Italy.
- Perspectives on regulation adaptation in multi-agent systems: from agent to organization-centric and beyond, *WESAAC@BRASIC 2025*, Fortaleza, Brazil

- A Unified View on Regulation Management in Multi-Agent Systems, *COINE@AAMAS 2025*, Online.
- Engineering Inter-Agent Explainability in BDI Agents, *EXTRAAMAS@AAMAS 2025*, Online.
- A Normative Agent-Centric Approach to Regulate Manufacturing Process, *SeReCo Autumn Workshop 2024*, Karlsruhe, Germany.
- An Agent-Centric Perspective on Norm Enforcement and Sanctions, *WOA 2024*, Bard, Italy.
- An Agent-Centric Perspective on Norm Enforcement and Sanctions, *COINE@AAMAS2024*, Online.
- Self-adaptive Regulation Mechanisms for a Trustworthy and Sustainable Industry of the Future, *IMT 2024 Colloquium "Responsible Industry of Future"*, Gardanne, France.
- Towards a Multi-Level Explainability Framework for Engineering and Understanding BDI Agent System, *WOA 2023*, Rome, Italy

Publications

Journals

- **Elena Yan**, Samuele Burattini, Jomi F. Hübner, and Alessandro Ricci A Multi-Level Explainability Framework for Engineering and Understanding BDI Agents. *Autonomous Agents and Multi-Agent Systems* **39**, 9 (2025). doi.org/10.1007/s10458-025-09689-6.

Conference Proceedings

- **Elena Yan**, Luis G. Nardin, Alessandro Ricci, Samuele Burattini, Guillaume Muller, Jomi F. Hübner, Jaime S. Sichman, and Olivier Boissier. Perspectives on the Explanation of Organizations in Multi-Agent Systems (Blue Sky Ideas). *Coordination, Organizations, Institutions, Norms, and Ethics for Governance of Multi-Agent Systems COINE@AAMAS2026* (To Appear)
- Jomi F. Hübner, **Elena Yan**, Luis G. Nardin, Olivier Boissier. Anticipatory Thinking in Normative Contexts *Coordination, Organizations, Institutions, Norms, and Ethics for Governance of Multi-Agent Systems COINE@AAMAS2026* (To Appear)
- **Elena Yan**, Samuele Burattini, Luis G. Nardin, Jomi F. Hübner, Olivier Boissier, Jaime S. Sichman, and Alessandro Ricci. Exploiting the MAOP Approach for Multi-Level Explainability of Multi-Agent Systems. *Workshop on Engineering Multi-Agent Systems EMAS@AAMAS2026* (To Appear)
- **Elena Yan**, Luis G. Nardin, Olivier Boissier, and Jaime S. Sichman. Un modèle d'adaptation de la régulation de systèmes multi-agents. *Journée Francophone des Systèmes Multi-Agent. JFSMA2026* (To Appear)
- **Elena Yan**, Luis G. Nardin, Olivier Boissier, and Jaime S. Sichman. A Regulation Adaptation Model for Multi-Agent Systems. *28th European Conference on Artificial Intelligence ECAI-2025*, Oct 2025, Bologna, Italy. doi.org/10.3233/FAIA251245.
- Katharine Beaumont*, **Elena Yan***, and Rem Collier. Towards an Ontology for Uniform Representations of Agent State for Heterogeneous Inter-Agent Explanations *2nd Workshop on Hypermedia Multi-Agent Systems co-located with ECAI 2025*. ceur-ws.org/Vol-4084/paper1.pdf
- **Elena Yan**, Luis G. Nardin, Olivier Boissier, Jaime S. Sichman, and Jomi F. Hübner. Perspectives on regulation adaptation in multi-agent systems: from agent to organization centric and beyond. In *Proceedings of the 19th Workshop-School on Agents, Environments, and Applications*, September 29, 2025, Fortaleza/CE, Brasil. SBC, Porto Alegre, Brasil, 39-50. sol.sbc.org.br/index.php/wesaac/article/view/37527
- Katharine Beaumont, **Elena Yan**, Samuele Burattini, and Rem Collier. Agent Toolkits: Explainable Agency in ASTRA. *European Conference on Multi-Agent Systems EUMAS 2025* (To Appear).
- **Elena Yan**, Luis G. Nardin, Olivier Boissier, and Jaime S. Sichman. A Unified View on Regulation Management in Multi-Agent Systems. In: Tzeng, ST., Dell'Anna, D., Sichman, J.S., editors, *Coordination, Organizations, Institutions, Norms, and Ethics for Governance of Multi-Agent Systems XVIII*. COINE 2025. Lecture Notes in Computer Science(), vol 16253. Springer, Cham. doi.org/10.1007/978-3-032-17542-7_4
- Katharine Beaumont, **Elena Yan**, Samuele Burattini, and Rem Collier. Engineering Inter-Agent Explainability in BDI Agents. In: Calvaresi, D., et al. Explainable, Trustworthy, and Responsible AI and Multi-Agent Systems. EXTRAAMAS 2025. Lecture Notes in Computer Science(), vol 15936. Springer, Cham. doi.org/10.1007/978-3-032-01399-6_9
- **Elena Yan**, Luis G. Nardin, Jomi F. Hübner, and Olivier Boissier. An Agent-Centric Perspective on Norm Enforcement and Sanctions. In Stephen Cranefield, Luis G. Nardin, and Nathan Lloyd, editors, *Coordination,*

Organizations, Institutions, Norms, and Ethics for Governance of Multi-Agent Systems XVII, pages 79–99, Cham, 2025. Springer. doi.org/10.1007/978-3-031-82039-7_6.

- **Elena Yan**, Samuele Burattini, Jomi F. Hübner, and Alessandro Ricci. Towards a multi-level explainability framework for engineering and understanding BDI agent systems. In *Proceedings of the 24th Workshop "From Objects to Agents"*, Roma, Italy, November 6-8, 2023, Rino Falcone, Cristiano Castelfranchi, Alessandro Sapienza, and Filippo Cantucci, editors, CEUR Workshop Proceedings, vol. 3579, CEUR-WS.org, 2023, pp. 216–231. ceur-ws.org/Vol-3579/paper17.pdf.

Extended Abstracts

- **Elena Yan**, Samuele Burattini, Jomi F. Hübner, and Alessandro Ricci A Multi-Level Explainability Framework for Engineering and Understanding BDI Agents. *JAAMAS Track - Autonomous Agents and Multi-Agent Systems 2026*.
- **Elena Yan**. Self-Adaptive Regulation Mechanisms for a Trustworthy and Sustainable Industry of the Future. *ECAI 2025 - Doctoral Consortium*, Oct 2025, Bologna, Italy. ecai25doctoralconsortium.github.io/papers/ECAI-2025-DC_paper_45.pdf

*Authors marked with * indicate equal contribution.*

Skills

Languages: Chinese (Native/Bilingual), Italian (Native/Bilingual), English (Advanced), French (Intermediary), Portuguese (Basic)

Programming Languages: Java, Kotlin, Scala, JavaScript, Python, C++, C, C#, Prolog, Lua.

Technologies and Frameworks: JaCaMo, Android, Arduino, Node.js, React, Angular, Vue.js, Gradle, Docker, MongoDB, Git

Programming Paradigms: Multi-Agent Oriented Programming, Object Oriented Programming, Functional Programming, Logic Programming, Event Driven Programming, Imperative/Procedural Programming

Honors and Awards

Among the 3 Best Papers – 19th Workshop-School on Agents, Environments, and Applications WESAAC 2025. 2025

Best Paper Award – 24th Workshop "From Objects to Agents" WOA 2023. 2023

First Prize - Competition for the design of the logo for the various fields of application of the Municipality of Forlì. 2018

First Prize - National Art Prize "Iris Versari" 2nd Edition 2018